

CS211 Quiz 3 Review - Spring 2026

Example Database Schema

Here is a simple 3-table database for a school enrollment system:

Table 1: Students

student_id	name	email
@1234	Alice Johnson	alice@fitnyc.edu
@0234	Bob Smith	bob@fitnyc.edu
@0345	Carol White	carol@fitnyc.edu

Table 2: Courses

course_id	title	instructor
FM101	Retail Management	Dr. Brown
FM102	Inventory & Supply Chain	Dr. Green
FM103	Visual Merchandising	Dr. Blue

Table 3: Enrollments

enrollment_id	student_id	course_id	grade
1001	@1234	FM101	A
1002	@1234	FM102	B+
1003	@0234	FM101	A-
1004	@0234	FM103	B
1005	@0345	FM102	A

Quiz Questions

Question 1: What is the Model Context Protocol (MCP)?

- A) A protocol that allows LLMs to be trained on new data without retraining
 - B) A standard that enables LLMs to safely connect to and interact with external data sources and tools
 - C) A method for compressing the context window to make LLMs faster
 - D) A security framework that prevents LLMs from accessing any external systems
-

Question 2: Which of the following is a primary benefit of using tools with an LLM?

- A) Tools allow the LLM to permanently update its training data
 - B) Tools eliminate the need for a system prompt
 - C) Tools allow the LLM to perform actions beyond generating text, such as reading files, querying databases, or calling APIs
 - D) Tools reduce the cost of using an LLM by 50%
-

Question 3: In a database, what is a primary key?

- A) The first column in a table
 - B) A unique identifier that guarantees each row in a table can be uniquely identified
 - C) The key used to encrypt sensitive data in a database
 - D) A key that can contain null values to represent missing data
-

Question 4: What is a foreign key in a relational database?

- A) A key stored outside the database on a physical key drive
 - B) A column that references the primary key of another table, creating a relationship between tables
 - C) A backup key kept in a separate database for security
 - D) A key that is only accessible to users from other countries
-

Question 5: What is OpenCode in the context of LLM tools?

- A) An open-source programming language for building LLM applications
 - B) An agentic LLM harness that allows LLMs to interact with and manipulate files, folders, and other system resources
 - C) A database system that stores only publicly available code
 - D) A method for making LLM prompts visible to end users
-

Question 6: You want an LLM to fetch the current weather and then suggest what clothes to wear. Which of the following would you need?

- A) A chain of thought prompt only
 - B) A tool that connects to a weather API and a tool to access the user's closet
 - C) A tool that can call a weather API to retrieve current conditions
 - D) A system prompt that contains all possible weather conditions
-

Question 7: What is a connector?

- A) A physical cable that connects a computer to the internet
 - B) Another word for an MCP—a connection between an LLM and an external system or data source (like a database or file system)
 - C) A prompt that tells the LLM to connect two ideas together
 - D) A person who helps you debug LLM responses
-

Question 8: Looking at the Enrollments table above, why is it structured with 2 foreign keys instead of storing all data in one Students table?

- A) To make the database slower so users have time to read the data
 - B) To avoid duplicating course information and allow students to enroll in multiple courses without repeating course details
 - C) To prevent students from taking the same course twice
 - D) To require an LLM to join tables together before answering queries
-

Question 9: An LLM is connected to a tool that can query a database. A user asks: “What are the top 5 customers by total spending?” What should the LLM do?

- A) Make up realistic customer names and spending amounts
 - B) Use the database tool to query the actual data and return real results
 - C) Refuse to answer because the LLM doesn’t have direct database access
 - D) Ask the user to manually run a database query first
-

Question 10: What is the main advantage of using MCPs to connect an LLM to a database rather than embedding the entire database contents in the system prompt?

- A) MCPs are cheaper to use
 - B) MCPs allow the LLM to query only the data it needs, keeping the context window free for conversation and reasoning
 - C) MCPs prevent the LLM from ever making mistakes
 - D) MCPs allow the database to update the LLM’s training data automatically
-

Question 11: You are building a system where users can ask an LLM questions about their personal files (documents, spreadsheets, etc.). Which technology would you need?

- A) A system prompt that contains all possible file paths
 - B) An MCP connector to the user’s file system and a tool that can read and analyze files
 - C) A foreign key relationship between the user and their files
 - D) A chain of thought prompt that lists every file in the system
-

Question 12: In the example database above, if you wanted to find “all courses that Alice is enrolled in”, you would need to:

- A) Look only in the Students table
 - B) Look only in the Courses table
 - C) Query the Enrollments table using her student_id, then join with the Courses table to get course names
 - D) Ask the LLM to guess based on her name
-

Question 13: What should happen if you try to insert a record with a foreign key value that doesn't exist in the referenced table?

- A) The database should silently ignore the foreign key
 - B) The database should automatically create a matching record in the referenced table
 - C) The database should reject the insertion to maintain data integrity
 - D) The LLM should override the database constraint
-

Question 14: An LLM is using OpenCode to help a user debug their Python script. What can OpenCode do that a regular chat interface cannot?

- A) Create a PDF file of the conversation
 - B) Execute the code in a sandboxed environment and show the output or error messages in real-time
 - C) Train the LLM on the user's code
 - D) Automatically upload the code to GitHub
-

Question 15: A company wants to give an LLM access to their customer database so it can answer customer service questions like "What was my last order?" Which of the following is the most secure approach?

- A) Give the LLM the database admin password
 - B) Copy the entire customer database into the LLM's system prompt
 - C) Use an MCP with a connector that allows the LLM to query only specific, pre-approved data and cannot delete or modify records
 - D) Disable all security checks so the LLM can access the database freely
-

Question 16: An AI App sends a user prompt and a list of available tools to an LLM. Describe in your own words what happens next. Include: what the LLM decides, how the tool gets called, what happens after the tool runs, and how the final answer reaches the user.

Answer Key

1. B
2. C
3. B
4. B
5. B
6. C
7. B
8. B
9. B
10. B
11. B
12. C
13. C
14. B
15. C